



Molar Pregnancy & Gestational Trophoblastic Disease

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**COLLEGE OF
MEDICINE**

Objectives

- Identify risk factors for molar pregnancy
- Identify risk factors for gestational trophoblastic disease
- Describe initial workup and management for patients with molar pregnancy
- Describe initial workup and management for patients with gestational trophoblastic disease
- Discuss pathology and pathophysiology of molar pregnancy
- Discuss pathology and pathophysiology of gestational trophoblastic disease



Case Presentation

A 16-year-old G1P0 presents to the hospital with vaginal bleeding and passage of tissue from the vagina. Her LMP was about 14 weeks ago. She has been experiencing nausea and vomiting. She has not yet established prenatal care.

She has no prior medical or surgical history. Family history is unremarkable.

On examination, her heart rate is 115. Her fundus is palpable on examination to the level of the umbilicus. Pelvic exam is pertinent for vesicular tissue in the vaginal fornix along with a scant amount of vaginal bleeding. Nursing is unable to obtain fetal heart tones.

Laboratory testing reveals hCG level of 478,000.



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Laboratory testing reveals **hCG level of 478,000**.

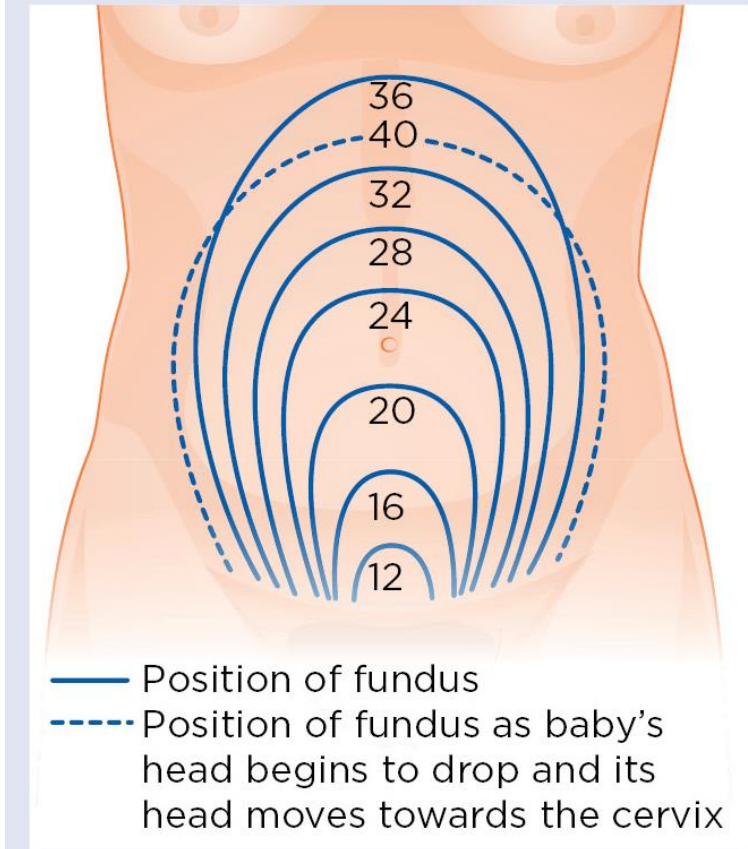


HCG Levels Chart

WEEKS FROM LAST MENSTRUAL PERIOD	APPROX. AMOUNT OF HCG (IN MIU/ML)
3 weeks	5 to 50
4 weeks	5 to 426
5 weeks	19 to 7,340
6 weeks	1,080 to 56,500
7-8 weeks	7,650 to 229,000
9-12 weeks	25,700 to 288,000
13-16 weeks	3,300 to 253,000
17-24 weeks (second trimester)	4,060 to 165,400
25 to term (third trimester)	3,640 to 117,000
After several days postpartum	Nonpregnant levels (<5)

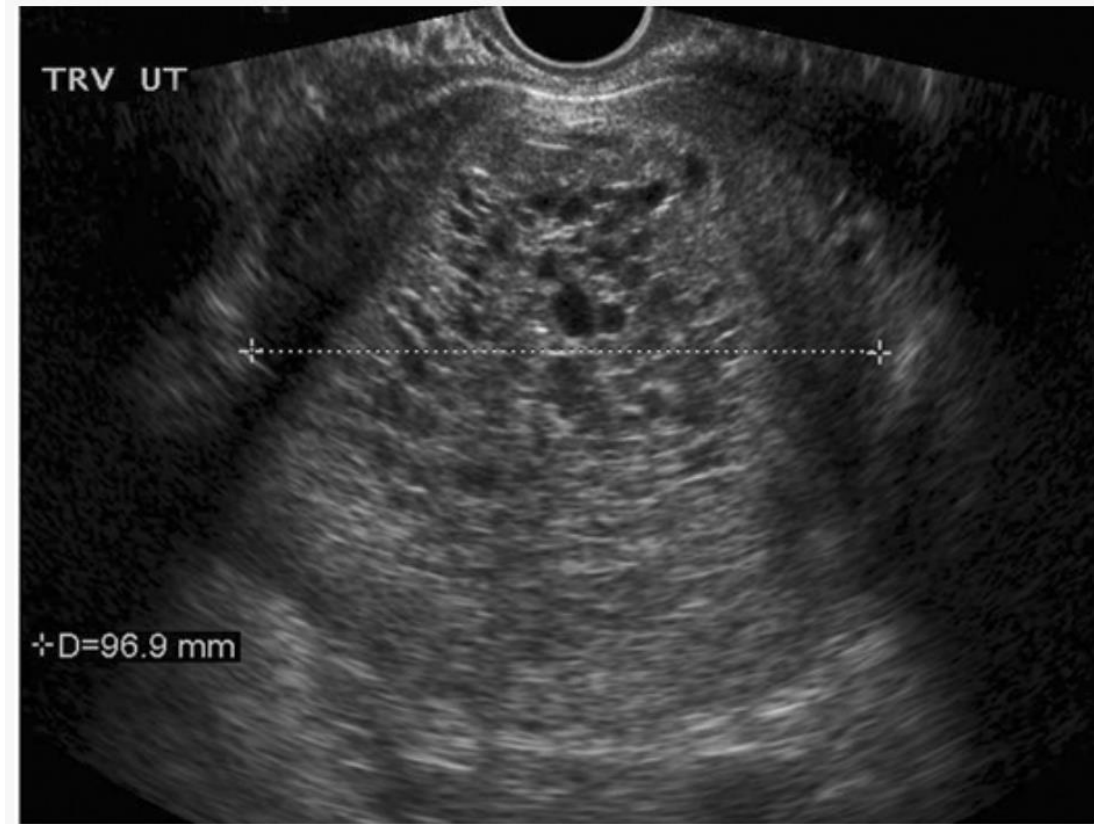
what to expect.

Fig 4. **Changes to uterine size and position**



Molar Pregnancy

- Classic presentation:
 - Second trimester passage of grape-like tissue
 - No fetal heart tones
 - S>D
- With prenatal care
 - “Snowstorm appearance” on ultrasound
 - Elevated hCG



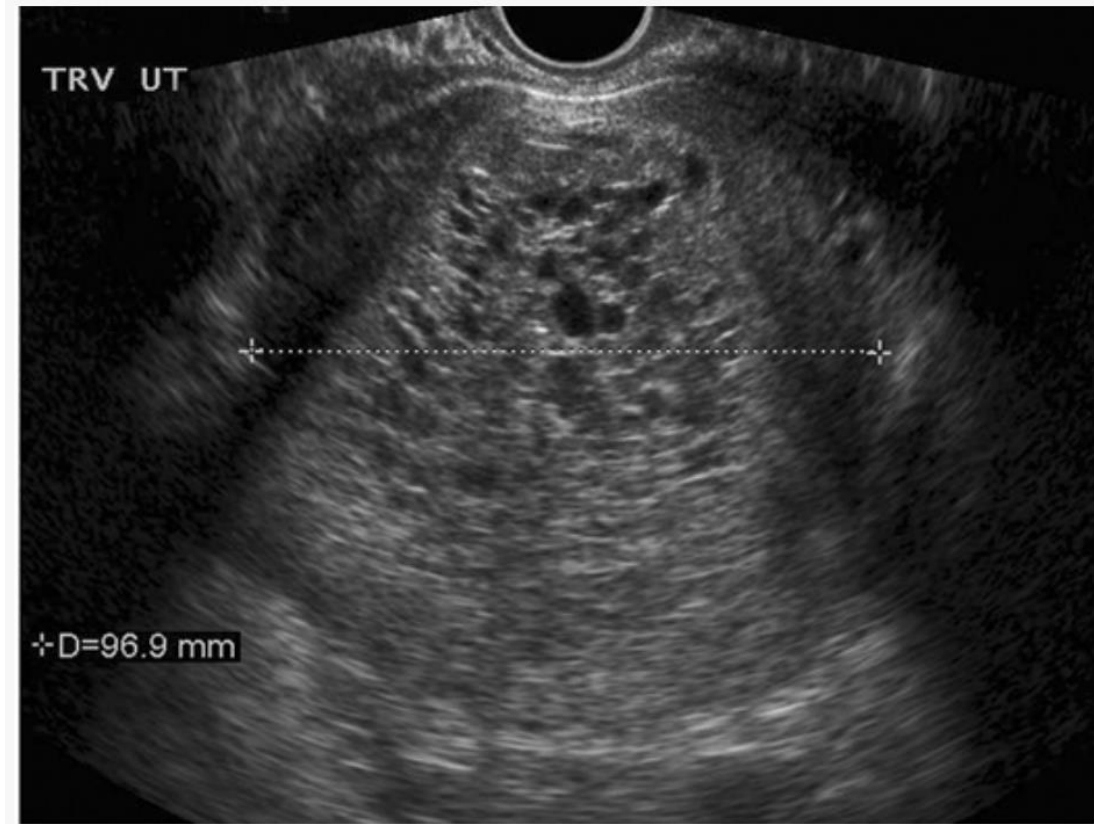
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FIGURE 37-5

Transverse sonographic view of a uterus with a complete hydatidiform mole. The classic “snowstorm” appearance is created by the multiple placental vesicles. The mole completely fills this uterine cavity, and calipers are placed on the outer uterine borders.

Molar Pregnancy

- May present to the ED with:
 - Early pre-eclampsia
 - Thyroid storm
 - Hyperemesis gravidarum
 - What hormone is responsible?

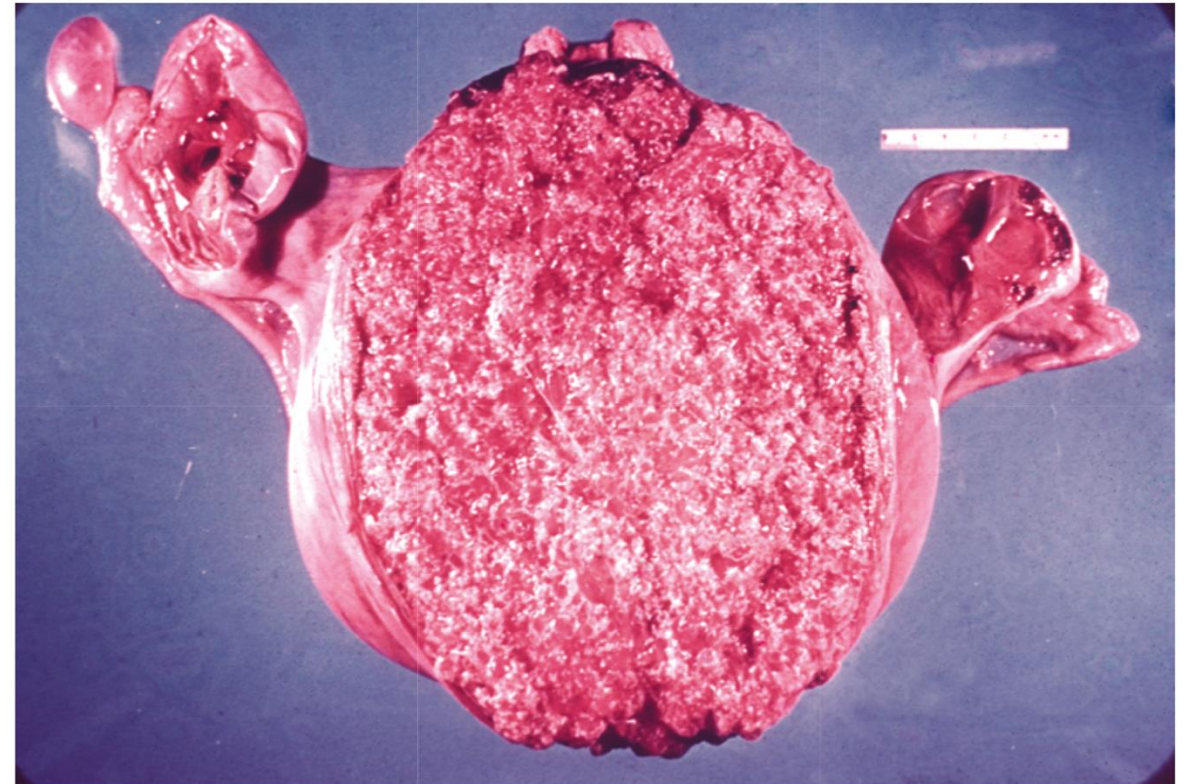
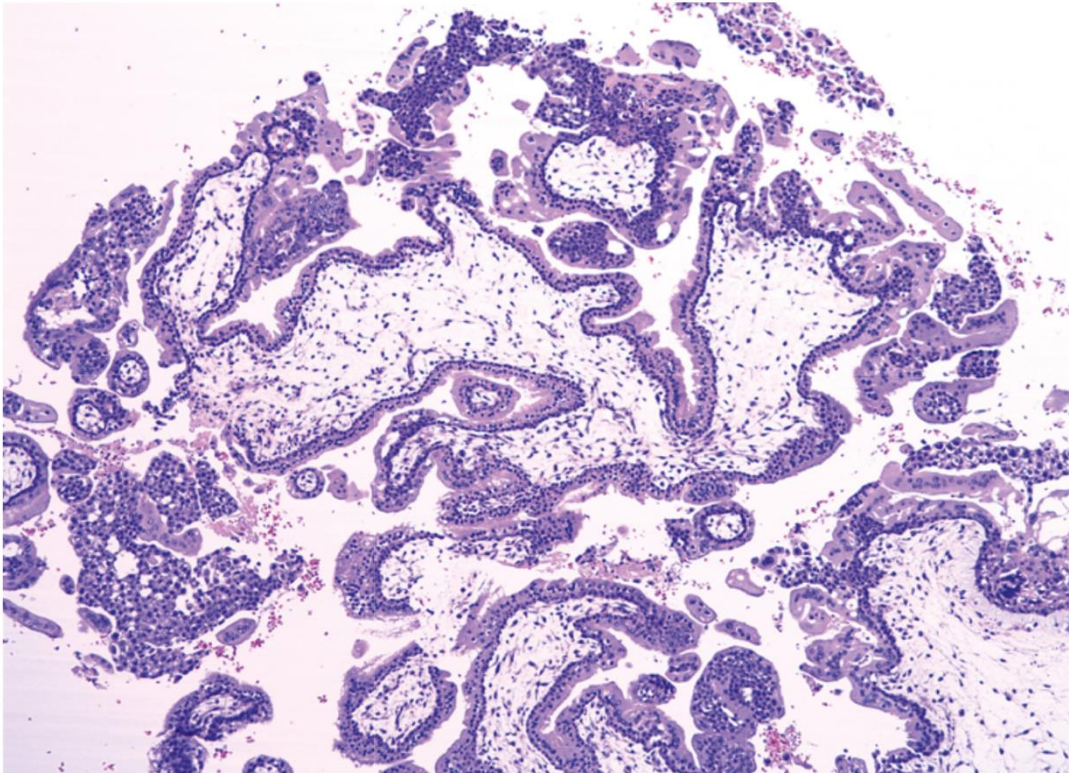


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FIGURE 37-5

Transverse sonographic view of a uterus with a complete hydatidiform mole. The classic “snowstorm” appearance is created by the multiple placental vesicles. The mole completely fills this uterine cavity, and calipers are placed on the outer uterine borders.

Molar Pregnancy



Molar pregnancy: Risk Factors

- Extremes of age (<15 yo or >45 yo)
 - Risk is 5–10 times greater for > 45 yo
- History of prior molar pregnancy
 - Second molar pregnancy: risk 1–2% (vs 0.1%)
 - Third molar pregnancy: risk 15–20%
 - Risk is much higher after a complete mole
- Nulliparity
- Familial recurrent hydatidiform mole syndrome (FRHM)
 - Autosomal recessive *NLRP7* mutation
- Incidence: 1-2/1000 pregnancies in western countries
 - (2-3 times higher in Asia, Brazil, India)

Case Presentation

A 45-year-old G3P1011 presents to the hospital with vaginal bleeding. Her LMP was about 6 weeks ago. She has been experiencing nausea. She has not yet established prenatal care.

She has no prior medical or surgical history. Family history is unremarkable. She has a history of one prior spontaneous abortion and one prior full term vaginal delivery

On examination, her heart rate is 105. Pelvic exam is pertinent for a moderate amount of vaginal bleeding.

Laboratory testing reveals hCG level of 45,000.



Case Presentation

A **45-year-old** G3P1011 presents to the hospital with vaginal bleeding. Her LMP was about 6 weeks ago. She has been experiencing nausea. She has not yet established prenatal care.

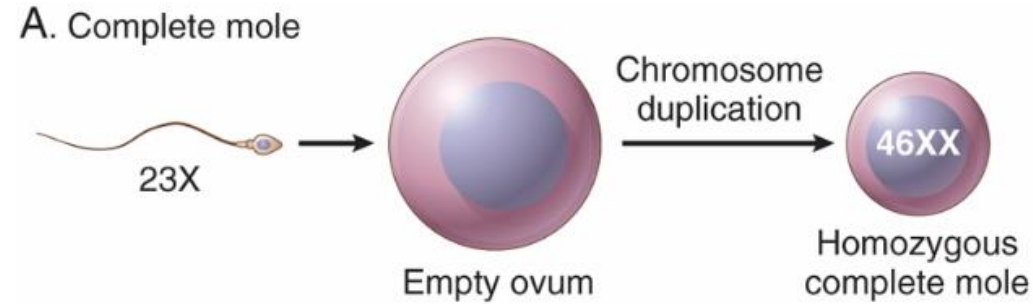
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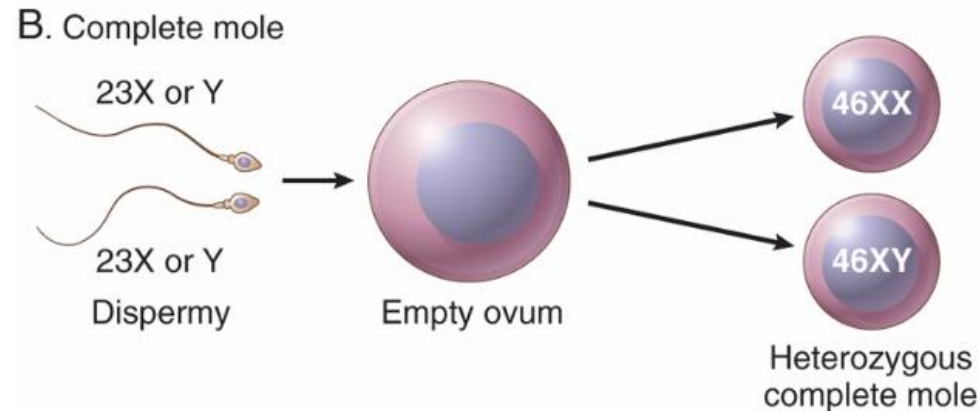
Laboratory testing reveals **hCG level of 45,000**.

Complete Mole: Genetics

More common

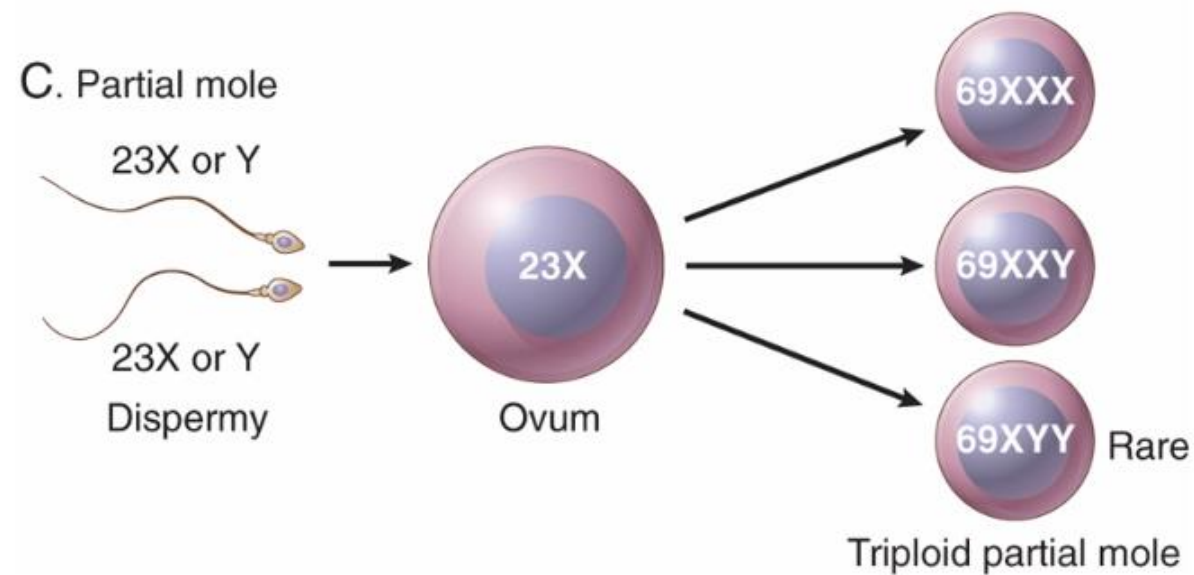


Less common



karyotypes are diploid and are completely paternally derived

Partial Mole: Genetics



Molar Pregnancy: Summary

	Partial Mole	Complete Mole
Pathophysiology	Normal ovum fertilized by two sperm (or one sperm duplicates chromosomes)	Empty ovum fertilized by two sperm (or one sperm duplicates chromosomes)
Karyotype	69XXY, 69XXX, or 69XYY	46XX or 46XY
Fetal Parts?	Present	Absent
hCG	Elevated or normal	Super elevated
Presentation	Suspected missed abortion	Suspected mole
Invasive Mole	4-6% risk	15% risk
Choriocarcinoma	Minimal risk	2% risk



Molar Pregnancy: Summary

	Partial Mole	Complete Mole
Pathophysiology	Normal ovum fertilized by two sperm (or one sperm duplicates chromosomes)	Empty ovum fertilized by two sperm (or one sperm duplicates chromosomes)
Karyotype	69XXY, 69XXX, or 69XYY	46,XX,XXY,XXYY,XXYY
Fetal Parts?	Present	Absent
hCG	Elevated or normal	Significantly elevated
Presentation	Suspected missed abortion	Suspected molar pregnancy
Invasive Mole	4-6% risk	15-20% risk
Choriocarcinoma	Minimal risk	20-30% risk

Easy mnemonic:
Partially male
Part baby
Partially elevated hCG
Part risk for cancer



Molar Pregnancy: Summary

	Partial Mole	Complete Mole
Pathophysiology	Easy mnemonic: Completely male Completely abnormal tissue (no baby) Completely elevated hCG Complete risk for cancer (sort of) Completely classic presentation	Empty ovum fertilized by two sperm (or one sperm duplicates chromosomes)
Karyotype		46XX or 46XY
Fetal Parts?		Absent
hCG		Super elevated
Presentation		Suspected mole
Invasive Mole		15% risk
Choriocarcinoma	Minimal risk	2% risk



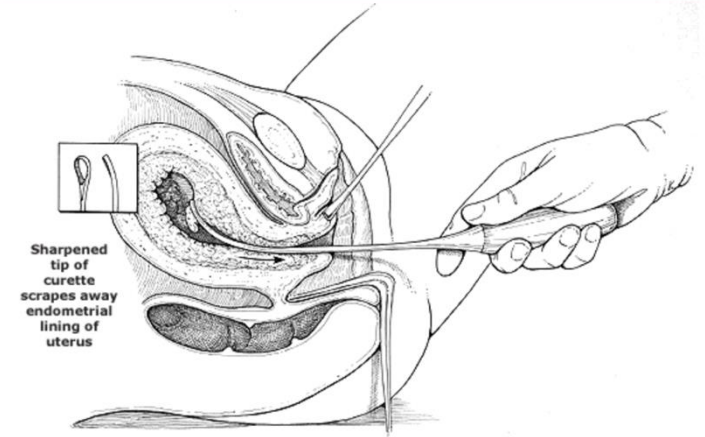
Molar Pregnancy: Summary

	Partial Mole	Complete Mole
Pathophysiology	You may hear the mnemonic “daddy’s little girl” - Usually 46 XX karyotype - Completely paternal genetics	Empty ovum fertilized by two sperm (or one sperm duplicates chromosomes)
Karyotype		46XX or 46XY
Fetal Parts?		Absent
hCG		Super elevated
Presentation		Suspected mole
Invasive Mole		15% risk
Choriocarcinoma		2% risk

Molar Pregnancy: Treatment 101

- Labs
 - Type & cross blood products (prepare for hemorrhage!)
 - Coags, Rh status, LFTs, CBC, creatinine
- Suction dilation and curettage with LARGE canula, uterotonics, +/- ultrasonographic guidance

Sharp curette inserted into uterine cavity



Courtesy of William J Mann, Jr, MD.

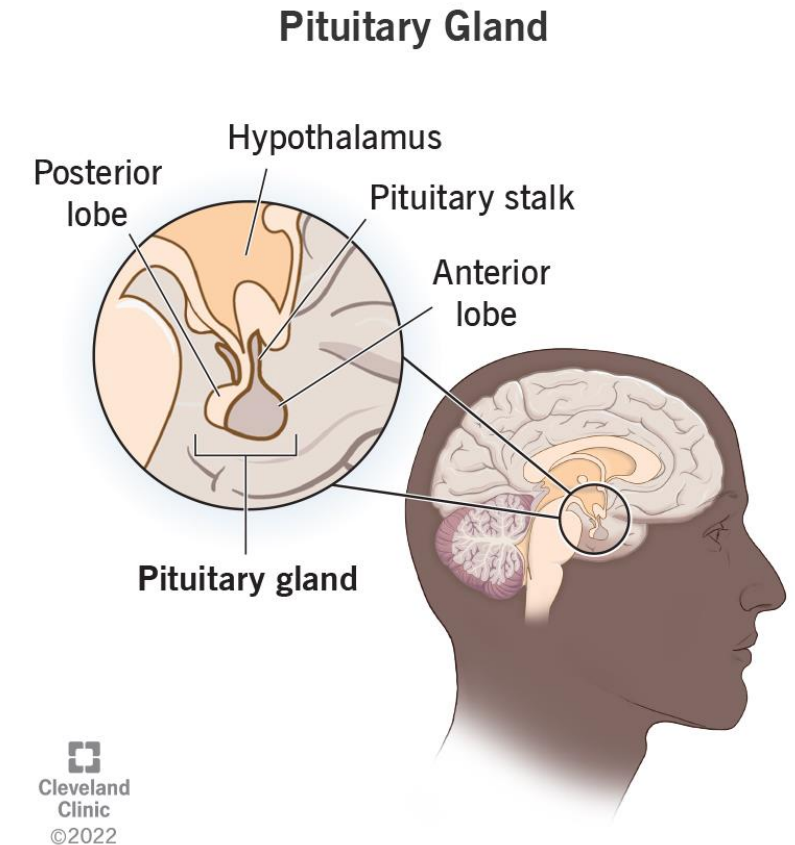


Molar Pregnancy: Treatment 101

- Medical management (e.g., misoprostol, mifepristone) ineffective, risks hemorrhage, and progression to GTN requiring chemotherapy
- Simple hysterectomy is alternative if fertility is not desired (especially if age >40yo)
- RhoGam if needed
- Contraception – what kind? Why is this critical?
- Follow hCG levels to zero and trend for ~6 months given risk of progression to invasive disease / choriocarcinoma
- Managed by an OB/GYN

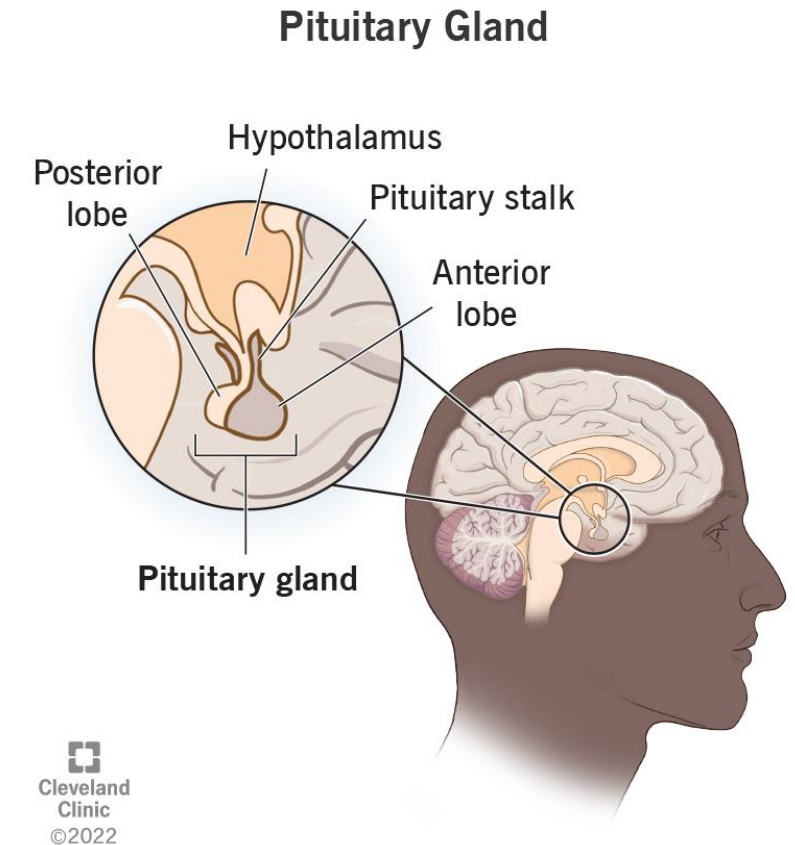
Beta Human chorionic gonadotropin (hCG)

- hCG: alpha and beta subunits
- Alpha subunit common to other pituitary hormones
- Beta subunit specific, so beta subunit is measured
- What are other pituitary hormones?



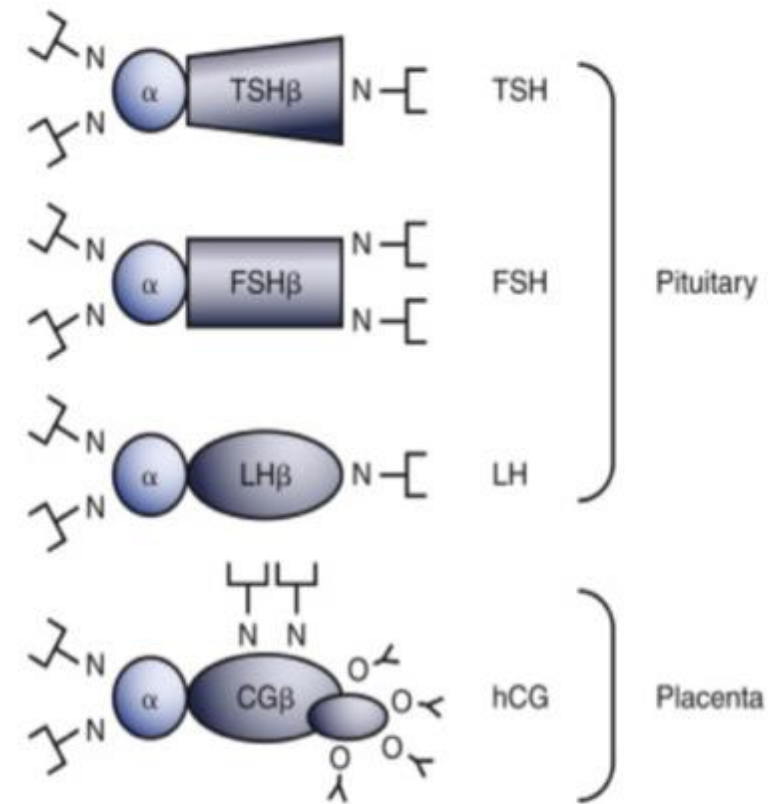
Pituitary hormones

- Anterior Pituitary Hormones
 - ACTH
 - FSH
 - GH
 - LH
 - PRL
 - TSH
- Posterior Pituitary hormones
 - ADH
 - Oxytocin



Human glycoprotein hormones

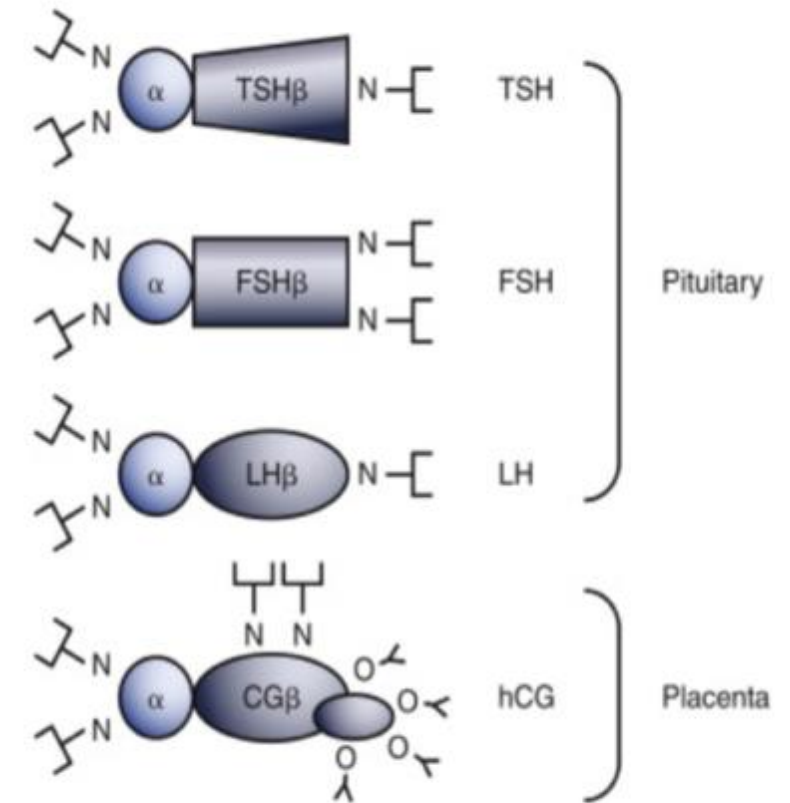
- Common alpha subunit?
- FSH
- LH
- TSH
- hCG



<https://quizlet.com/gb/491811682/theme-2-19-physiology-of-pregnancy-flash-cards/>

Beta hCG and TSH

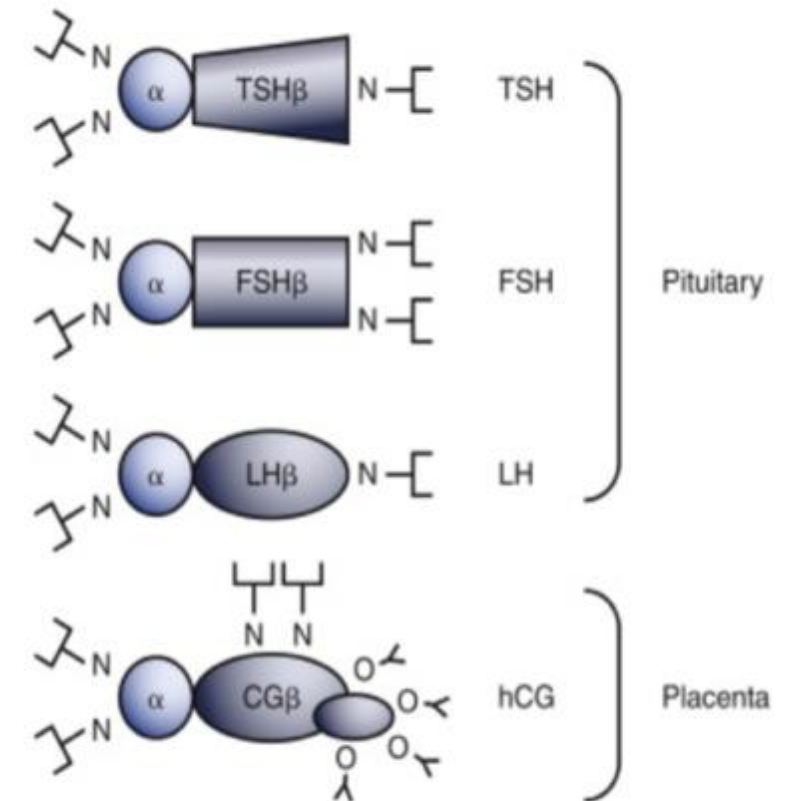
- Because of the super elevated level of beta hCG in complete molar pregnancy, this can mimic TSH, causing increased T4
- This causes negative feedback to decrease TSH
- Can present with thyroid storm
- What are signs and symptoms of thyroid storm?



<https://quizlet.com/gb/491811682/theme-2-19-physiology-of-pregnancy-flash-cards/>

Beta hCG and TSH

- Elevated circulating levels of hCG seen in complete molar pregnancy can increase T4 secretion, which, in turn, suppresses TSH.
- hCG and TSH receptors exhibit similar homology of approximately 70% for the transmembrane domain and 45% for the extracellular domains of the receptors.
- Hyperthyroidism is known to be associated with GTDs because of this cross-reactivity and significantly higher and sustained β -hCG levels compared to normal pregnancy.



<https://quizlet.com/gb/491811682/theme-2-19-physiology-of-pregnancy-flash-cards/>

Gestational Trophoblastic Disease: Epidemiology

- 80% of molar pregnancies do not progress to invasive disease
 - Most go on to have normal subsequent pregnancies without cancer
- 20% progress to invasive disease:
 - Locally invasive
 - Metastatic
- Molar Pregnancy
- Baseline incidence = **0.001%**
- Risk of recurrence after 1 mole = **1%**
- Risk of recurrence after 2 subsequent moles = **23%**



Malignant Gestational Trophoblastic Disease

- Managed by a gynecologic oncologist
- Three types of malignant histology:
 - Persistent / invasive moles (75%)
 - Choriocarcinoma (25%)
 - Placental Site Trophoblastic Tumor (<1%, extremely rare)
 - Epithelioid trophoblastic tumor (ETT) (<1%, even rarer)

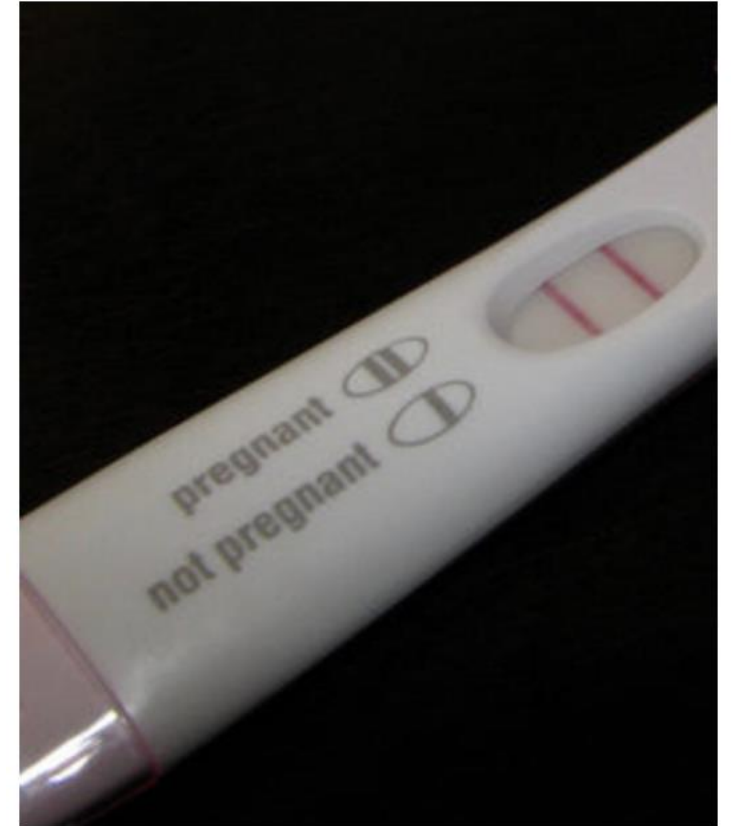
Who Gets Invasive Disease?

- Of those who have malignant GTD...
 - 50% had an antecedent molar pregnancy



- 25% had an antecedent normal pregnancy
- 25% had an antecedent abortion or ectopic pregnancy

Why is there a picture of a positive pregnancy test on this slide?

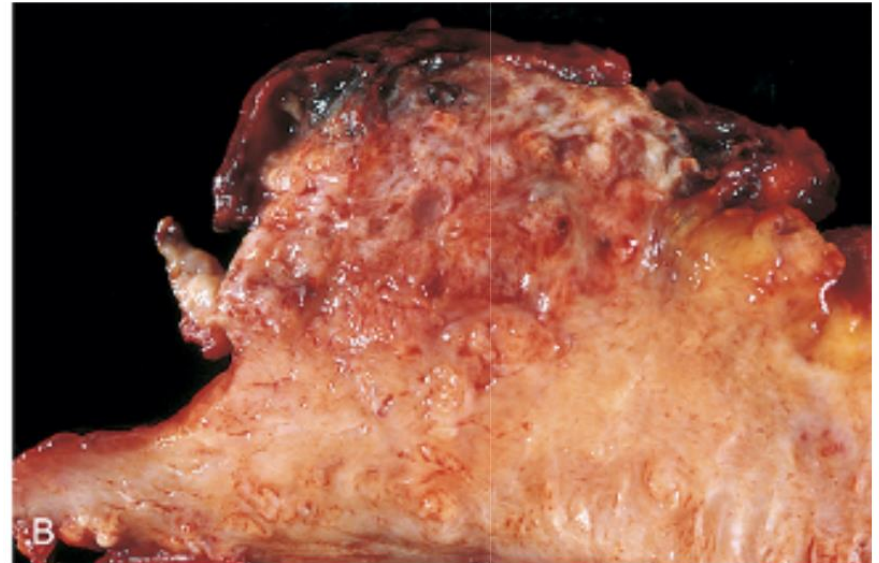


Invasive Mole

- Most commonly arises after a molar pregnancy
- Presentation:
 - Plateauing or rising hCG after a molar pregnancy;
 - May present as abnormal uterine bleeding
- Can spontaneously regress
- Rarely metastasize
- Incidence 1/15,000 pregnancies

Treatment: chemotherapy (methotrexate)
Fertility can be spared

Where else have you seen methotrexate used?



*Locally invasive
Hydropic villi and
trophoblasts invade
the myometrium*

Methotrexate

- How does methotrexate work?
 - Folate antimetabolite that inhibits DNA synthesis, repair, and cellular replication.
 - Binds to and inhibits dihydrofolate reductase
 - Inhibits formation of purine and thymidylic acid synthesis
 - Interferes with DNA synthesis, repair, and cellular replication.
 - Methotrexate is cell cycle specific for the S phase of the cycle.
 - Actively proliferative tissues are more susceptible to the effects of methotrexate.



<https://www.medicalnewstoday.com/articles/325549>

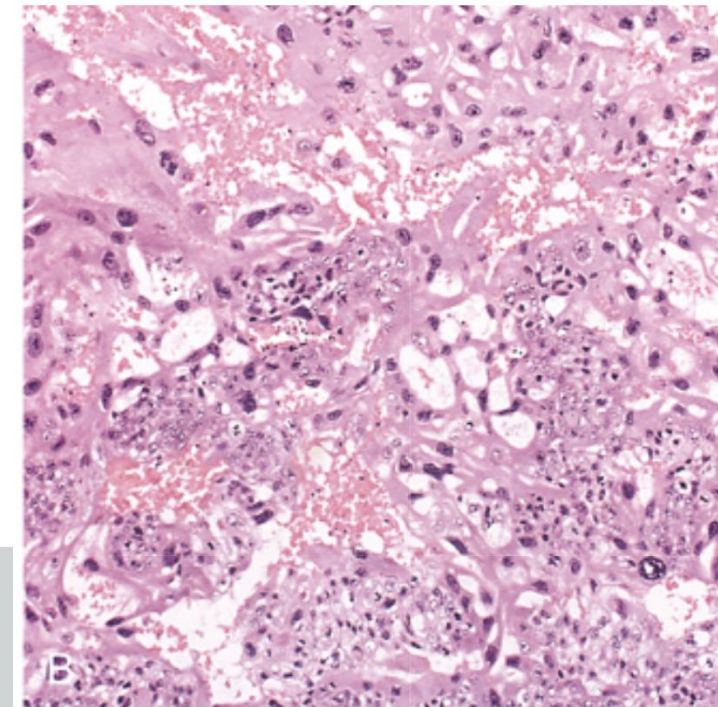
Choriocarcinoma

- Hematogenous spread, **vascular invasion**
 - Lungs, Vagina, Pelvis
 - Brain, Liver, Intestines
 - A rare metastatic cancer that can be cured
- **Elevated beta hCG (tumor marker)**
- Arises after a normal pregnancy, abortion, or ectopic pregnancy
- Presentation: **late post-partum bleeding**
- If arises from gestational pathway (this one!)
 - Chemo-sensitive
- Ovarian germ cell pathway
 - Chemo-resistant
- Treated with multi-agent chemotherapy; fertility spared

Grossly hemorrhagic
Uterine invasion
Risk of hemorrhage



Mimics placenta
No villi!

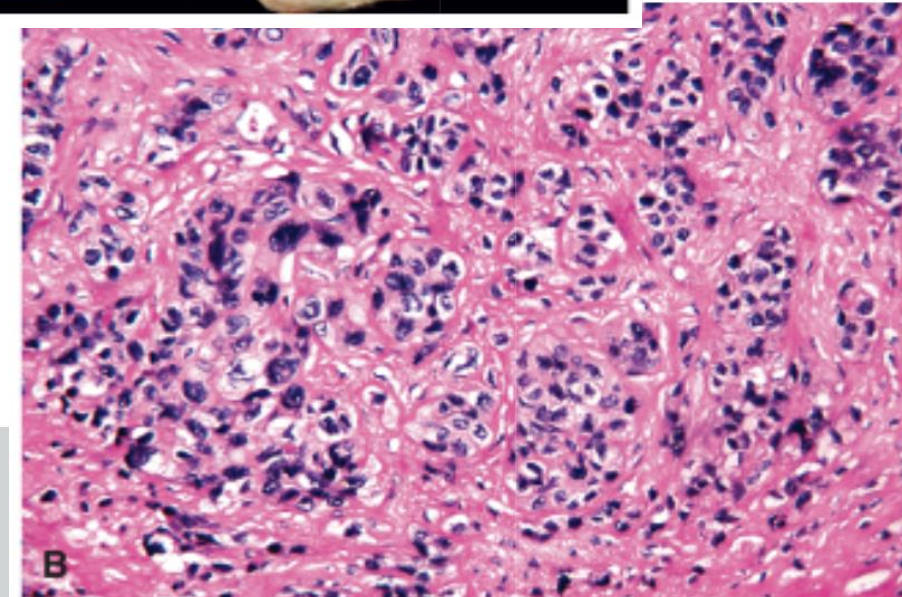
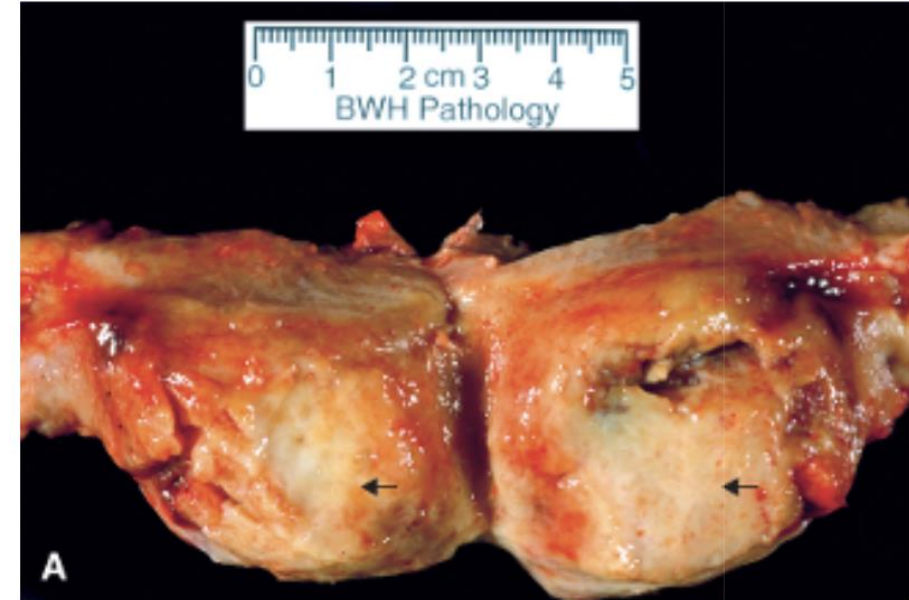


Side-note: Non-Gestational Choriocarcinoma

- Extremely rare
- Germ cell tumor (ovaries and testes)
- Can't distinguish gestational vs non-gestational on IHC or tumor markers-need tumoral genetics
- Ovarian: poor prognosis. Do not respond to chemo.
- You can also technically have a gestational choriocarcinoma of the ovary (metastasis or ovarian pregnancy)

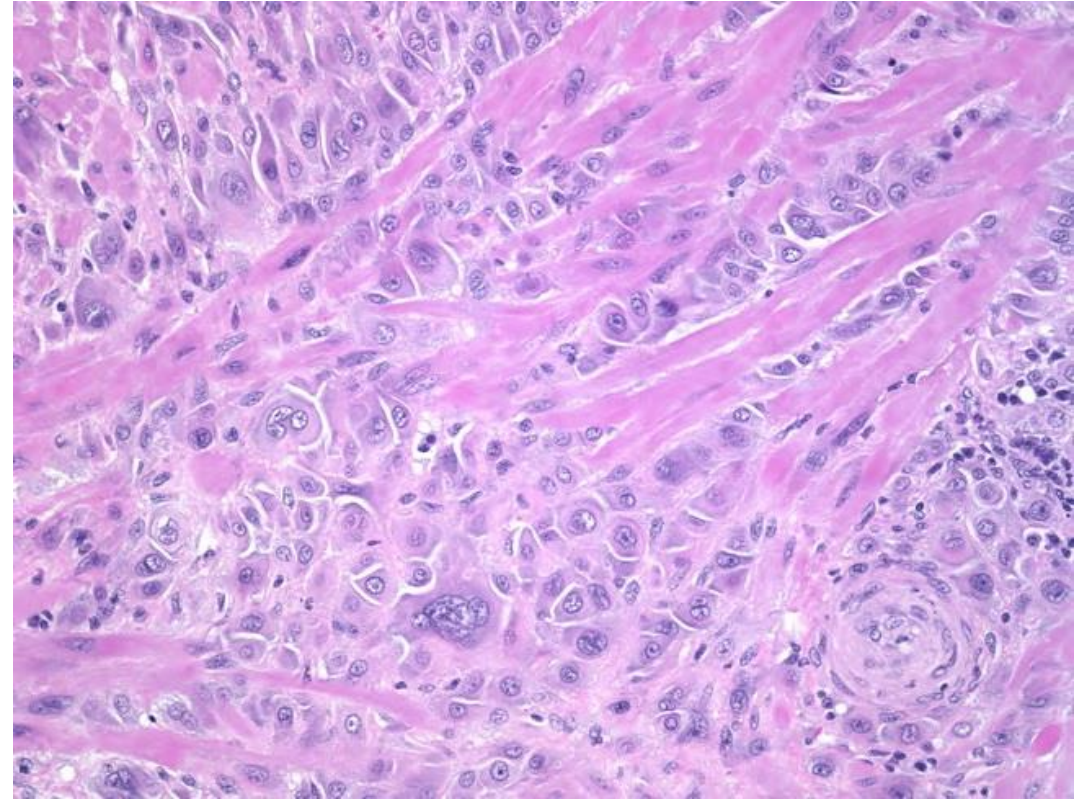
Placental-Site Trophoblastic Tumor (PSTT)

- Rare (<2%) of GTN
- Neoplasm of intermediate trophoblasts (normally found at the implantation site)
- Presentation:
 - Uterine mass
 - Abnormal uterine bleeding or amenorrhea
 - Moderately elevated hCG
- 85% have local disease (excellent prognosis)



Placental Site Trophoblastic Tumor (PSTT)

- Not usually post-molar
- Often diagnosed 18 mos after term pregnancy
- Slow growing
- Those with local disease are cured with hysterectomy +/- lymphadenectomy
- High risk disease may require adjuvant multi-agent chemo
- Metastases are rare, but treated with debulking and multiagent chemo
 - Less sensitive to chemo
- About 15% die of disease

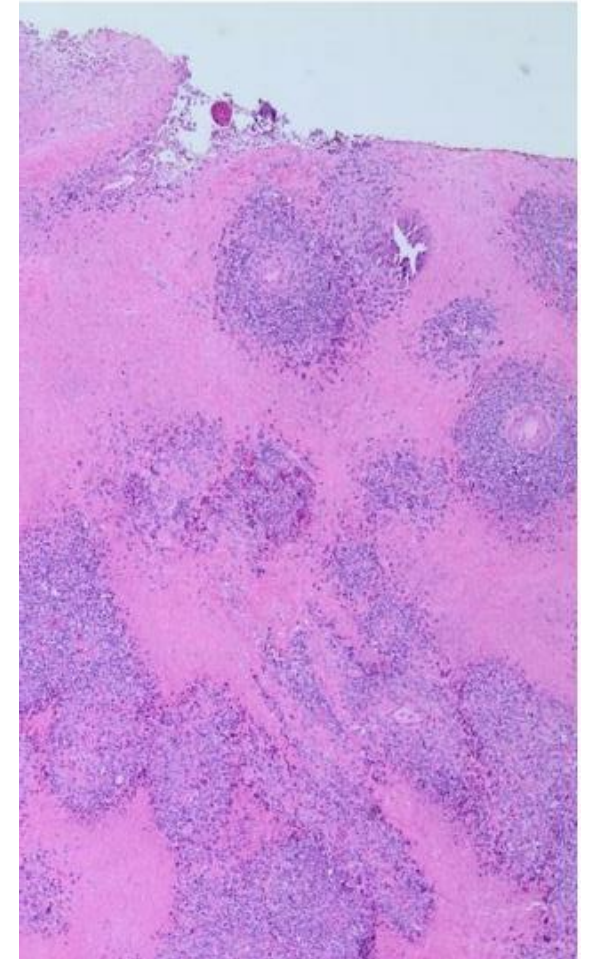


Epithelioid trophoblastic tumor (ETT)

- Extremely rare
 - Incidence: 0.1 per 100,000 deliveries
- +hCG (<3000 mIU/ml)
- Arise from chorionic-type intermediate trophoblast cells
- Not usually post-molar (avg 6 yrs post preg)

Pathologic Features:

- Nodular growth, deep invasion
 - Mild nuclear atypia
 - “geographic necrosis”
-
- Treated with surgery (hysterectomy) +/- multiagent chemotherapy
 - Metastasis in 25%
 - Less sensitive to chemo
 - Half die of disease



Questions?

